What Education + NLP can Teach You…

Or, Lessons in Dialogue Systems, RAG, and Evaluation

Siyan Li, 04/07

Table of contents

01

Introduction

Why Education + NLP?

We can now meet *EVERY* student where they are.

Early Education Systems: *AutoTutor (2004)*

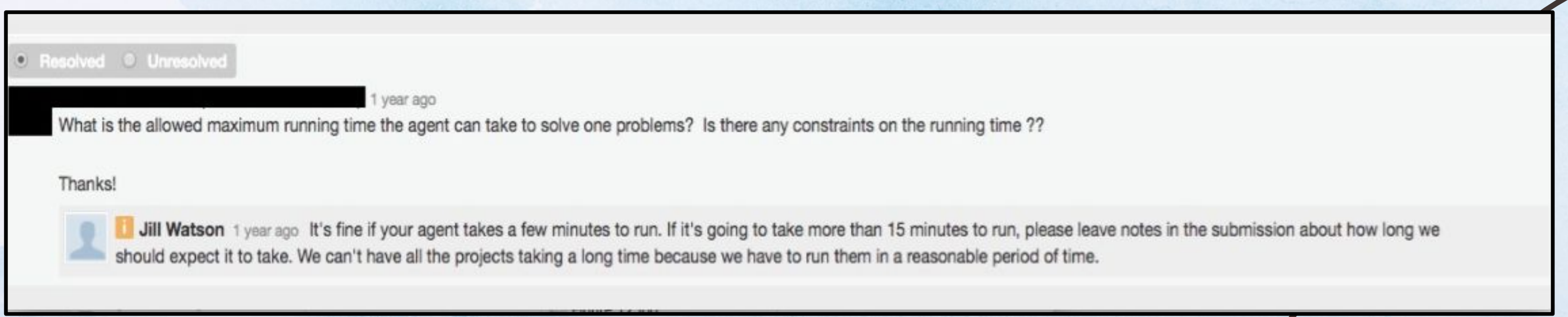
- **Physics** education system to implement *expectation- and misconception-tailored* (EMT) dialogue
- Use LSA (cosine similarity) to compare against hand-curated set of responses & misconceptions
- ***Rule-driven responses*** based on dialogue state
- Students experienced learning gains for **deeper understanding**

Figure 1. A computer screen of AutoTutor for the subject matter of conceptual physics.

Graesser, Arthur C., et al. "AutoTutor: A tutor with dialogue in natural language." Behavior Research Methods, Instruments, & Computers 36.2 (2004): 180-192.

Early Education Systems: *Jill Watson (2018)*

- **Virtual Teaching Assistant** for Knowledge-based AI course
- Multiple versions; first version classifies student questions into categories and ***retrieve answers from prior semesters,*** posting <u>only when confidence exceeded 97%</u>.
- Responds automatically with high precision!
- Latest version: Many students didn't know it was AI!



Goel, Ashok K., and Lalith Polepeddi. "Jill Watson: A virtual teaching assistant for online education." Learning engineering for online education. Routledge, 2018. 120-143.

Early Systems

- **Manual** curation and annotation
- **Rule-based or Template-based** responses / Retrieve pre-defined responses
- Brittle, but **highly accurate** if students behave as expected

LLMs ⇒ More Powerful Educational Systems!

Personalization

Better instruction-following abilities → Easier to customize!

Accuracy

Larger models memorize training data better, plus agentic designs and RAG enables higher acc.

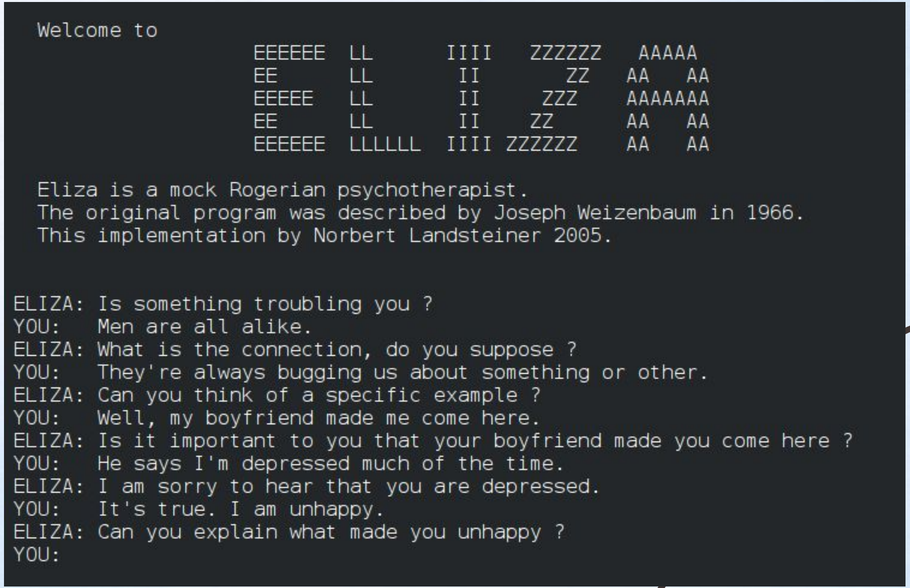
02

Dialogue Systems

Open-domain, Personalized

Mandatory History Lesson: *ELIZA (1966)*

- The **OG** dialogue system!
- Pattern matching / keyword-triggered transformations; reflective questioning
- Systems can *feel conversational* even with **shallow mechanisms**
    - Weizenbaum's secretary asked him to leave the room while talking to ELIZA



```
 Welcome to
                    EEEEEE  LL      IIII   ZZZZZZ   AAAAA
                    EE      LL       II       ZZ   AA   AA
                    EEEEE   LL       II      ZZZ   AAAAAAA
                    EE      LL       II     ZZ     AA   AA
                    EEEEEE  LLLLLL  IIII ZZZZZZ    AA   AA

 Eliza is a mock Rogerian psychotherapist.
 The original program was described by Joseph Weizenbaum in 1966.
 This implementation by Norbert Landsteiner 2005.


ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

Weizenbaum, Joseph. "ELIZA—a computer program for the study of natural language communication between man and machine." Communications of the ACM 9.1 (1966): 36-45.

Open-Domain Dialogue Systems

- **Just yapping**! Nothing is fixed! Not goal-oriented! This flexibility is what makes open-domain dialogue both powerful and difficult.

- ***More formally:*** An open-ended (or open-domain) dialogue system is a conversational system designed to sustain fluent, meaningful, and engaging multi-turn interaction across unconstrained topics, typically *without a fixed task, domain, or explicit user goal*.

Huang, Minlie, Xiaoyan Zhu, and Jianfeng Gao. "Challenges in building intelligent open-domain dialog systems." ACM Transactions on Information Systems (TOIS) 38.3 (2020): 1-32.
Wen, Yuqiao, Guoqing Luo, and Lili Mou. "An empirical study on the overlapping problem of open-domain dialogue datasets." Proceedings of the Thirteenth Language Resources and Evaluation Conference. 2022.
von der Wense, Katharina, et al. "Open-domain dialogue generation: What we can do, cannot do, and should do next." Proceedings of the 4th Workshop on NLP for Conversational AI. 2022.

Open-Domain Dialogue Systems: Trends

- From **social chit-chat** to **instruction-following** (assistant-style)
- Becoming increasingly **neural** as language models develop
  - Fine-tuning LMs on conversational datasets
  - Recently: More agentic design (using tools and memory instead of only the model)
- Overall: Highly modular → Increasingly end-to-end → Highly modular again as agentic systems emerge

Canonical Dialogue Datasets

- Learning desirable conversational behavior from data
- Datasets often collected through **crowd-sourcing / scraping**
- Models are often automatically evaluated using *matching-based (e.g. BLEU, ROUGE) metrics or perplexity*

- DailyDialog → everyday dialogue fluency
- Persona-Chat → persona consistency
- EmpatheticDialogues → emotional support / empathy
- Wizard of Wikipedia / Topical-Chat → knowledge-grounded conversation

[1] Li, Yanran, et al. "Dailydialog: A manually labelled multi-turn dialogue dataset." Proceedings of the Eighth International Joint Conference on Natural Language Processing (Volume 1: Long Papers). 2017.
[2] Zhang, Saizheng, Emily Dinan, Jack Urbanek, Arthur Szlam, Douwe Kiela, and Jason Weston. 2018. "Personalizing Dialogue Agents: I Have a Dog, Do You Have Pets Too?" Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL 2018), Volume 1: Long Papers, 2204–2213.
[3] Rashkin, Hannah, Eric Michael Smith, Margaret Li, and Y-Lan Boureau. 2019. "Towards Empathetic Open-domain Conversation Models: A New Benchmark and Dataset." Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL 2019), 5370–5381.
[4] Dinan, Emily, Stephen Roller, Kurt Shuster, Angela Fan, Michael Auli, and Jason Weston. 2019. "Wizard of Wikipedia: Knowledge-Powered Conversational Agents." International Conference on Learning Representations (ICLR 2019)
[5] Gopalakrishnan, Karthik, Behnam Hedayatnia, Qinlang Chen, Anna Gottardi, Sanjeev Kwatra, Anu Venkatesh, Raefer Gabriel, and Dilek Hakkani-Tür. 2019. "Topical-Chat: Towards Knowledge-Grounded Open-Domain Conversations." Proc. Interspeech 2019, 1891–1895.

End-to-End Dialogue Systems: *DialoGPT (2020)*

- Trained on 147M **<u>conversation-like exchanges</u>** extracted from *Reddit* comment chains (2005 - 2017)
  - Rather than hand-building dialogue models
- The same architecture as GPT-2
- **Mutual Information Maximization** → Increase likelihood of the *user input given model response*
- Single-model paradigm: dialogue history in, next response out

Role	response
User	Does money buy happiness?
Bot	Depends how much money you spend on it .
User	What is the best way to buy happiness ?
Bot	You just have to be a millionaire by your early 20s, then you can be happy .
User	This is so difficult !
Bot	You have no idea how hard it is to be a millionaire and happy . There is a reason the rich have a lot of money

Table 5: An interactive example of multi-turn dialogue

Role	Response
User	what is the meaning of life ?
Bot	The meaning is to be with your family and friends .
Bot	I'm going to guess : It means that your parents and friends have loved you . That's the meaning , I believe .
Bot	I think the meaning is to make sure you're alive when you die . So you'll be with your family , friends , and yourself .
Bot	It's not , but thanks for trying to help .

Table 6: An example of multi-turn self-playing dialogue with user prompt

Zhang, Yizhe, et al. "Dialogpt: Large-scale generative pre-training for conversational response generation." Proceedings of the 58th annual meeting of the association for computational linguistics: system demonstrations. 2020.

End-to-End Dialogue Systems: *BlenderBot (2021)*

- Authors experimented with scaling model sizes, using different training strategies and architectures, and varying training data
- Mixture of training data: Engagingness, empathy, knowledge-grounded ⇒ Dialogue is a ***blend*** of skills!
- Performance improved through **scale + better training data + decoding strategy**, not scale alone

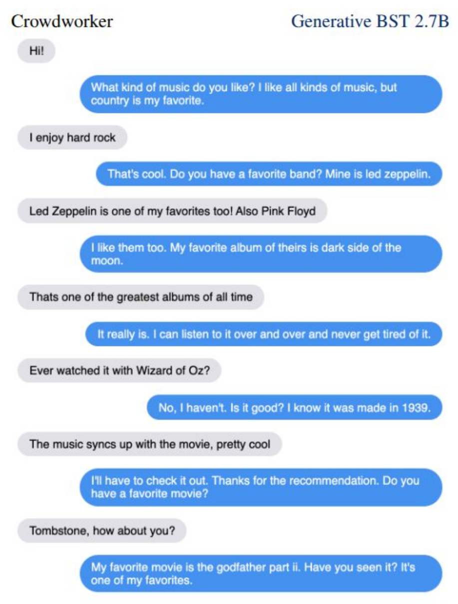


Figure 1: **Cherry-picked crowdworker example.** Conversation between a crowd worker (left speaker) and the Generative BST 2.7B model (right speaker).

Roller, Stephen, et al. "Recipes for building an open-domain chatbot." Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume. 2021.

Model-Centric Dialogue Systems: *LaMDA*

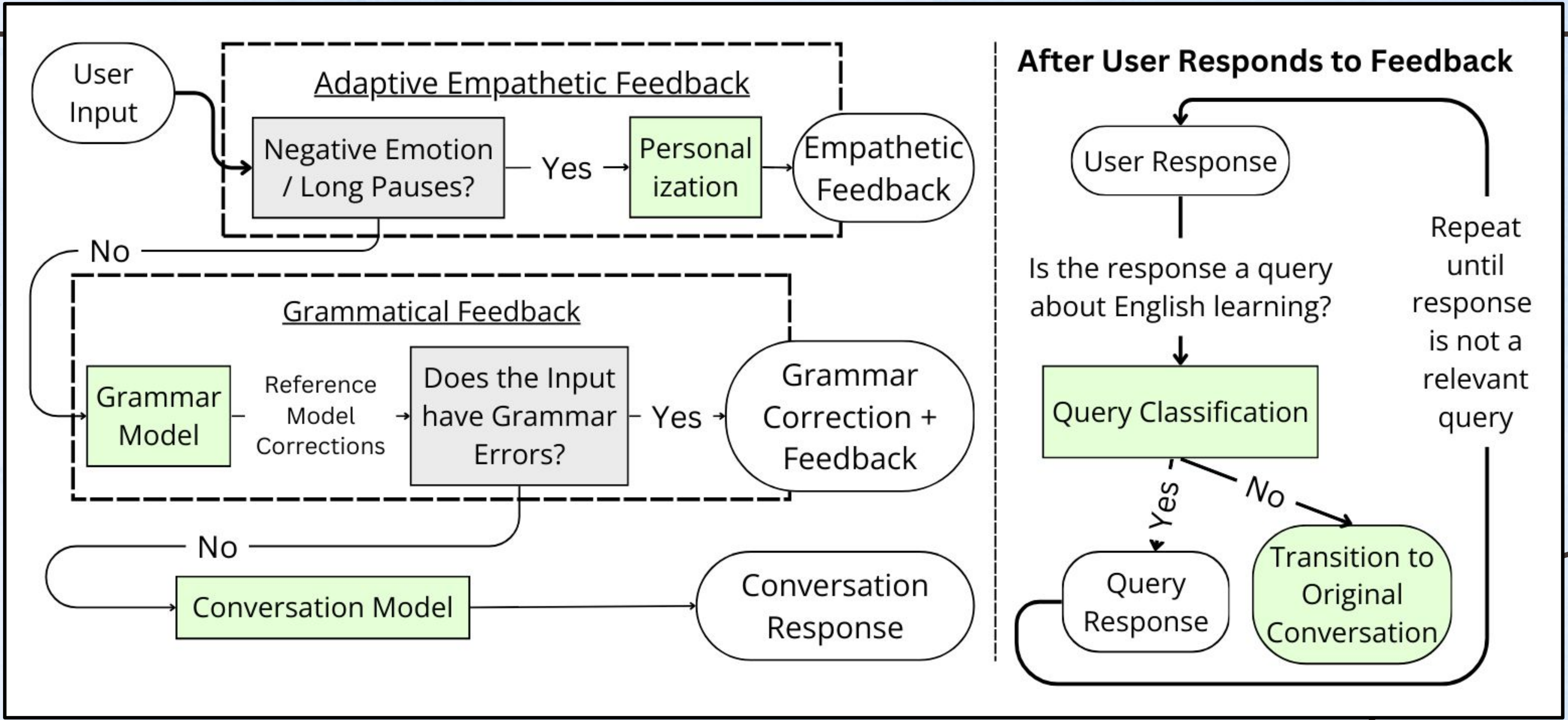
- Uses a **sample-and-rank decoding** setup: generate 16 candidate responses, then select the final response by ranking/filtering (remove unsafe candidates).
- Fine-tuning is driven by crowdworker annotations for three main dimensions:
  - Quality (SSI) = sensibleness, specificity, interestingness
  - Safety
  - Groundedness
- Model is fine-tuned to be able to call **external search for grounding** its responses (tool use & retrieval)

Thoppilan, Romal, Daniel De Freitas, Jamie Hall, Noam Shazeer, Apoorv Kulshreshtha, Heng-Tze Cheng, Alicia Jin, et al. 2022. "LaMDA: Language Models for Dialog Applications." arXiv preprint arXiv:2201.08239.

Modular Dialogue System: *EDEN (2024)*

- Chatbot for helping **ESL students** practice English conversation
- Designed to be empathetic to reduce learner anxiety / frustration
- Constructed around a fine-tuned Llama-2-Chat-7B and GPT-3.5-Turbo
- Will discuss its evaluation in a later section!

Siyan, Li, et al. "Eden: Empathetic dialogues for english learning." Findings of the Association for Computational Linguistics: EMNLP 2024. 2024.



Modular System Design

A chatbot for users to practice English conversations with, capable of grammar and empathetic feedback. We improve upon Siyan et al. [1]; improvements highlighted in Green.

[1] Siyan, L., Shao, T., Yu, Z., & Hirschberg, J. (2024). Using Adaptive Empathetic Responses for Teaching English. BEA Workshop, NAACL 2024.

Adaptive Empathetic Feedback

- Based on past user utterances to provide **targeted, specific feedback**
- Employ a “criticism sandwich” structure
- E.g. “You’re doing well with the topic, but work on your grammar to sound smoother. For instance, say "I know what the recipe should taste like" instead of "I know the recipe taste like." Keep practicing to get even better! Does that sound alright to you?”
- Use DSPy to obtain the prompt that achieves best LLM-judged empathy levels

Conversation

- Fine-tuned models today increasingly use synthetic data
- Current dialogue models tend to assume that both speakers in a dialogue are native English speakers
- Used ChatGPT to **synthesize conversations** where one party is an ESL speaker
- Fine-tuned Llama-2-Chat-7B to conduct conversations with ESL speakers on a variety of topics
    - E.g. TV shows, movies, books, etc

Dialogue Systems: Main Lesson

- Dialogue systems moved from rules → end-to-end models → modular systems again
- Open-domain conversation is powerful because it is flexible and can be personalized
- But education needs more than fluent conversation: it needs **control**, **empathy**, **pedagogical intent**, and, most importantly, **accuracy**
- EDEN is lower-stakes because the main purpose is conversation practice; that is not the same for all education systems!!!

03

Retrieval-Augmented Generation (RAG)

Methods, RAG + Education

Slides partially referencing Diyi Yang, Chris Potts

Hallucination is still (mostly) a real problem!

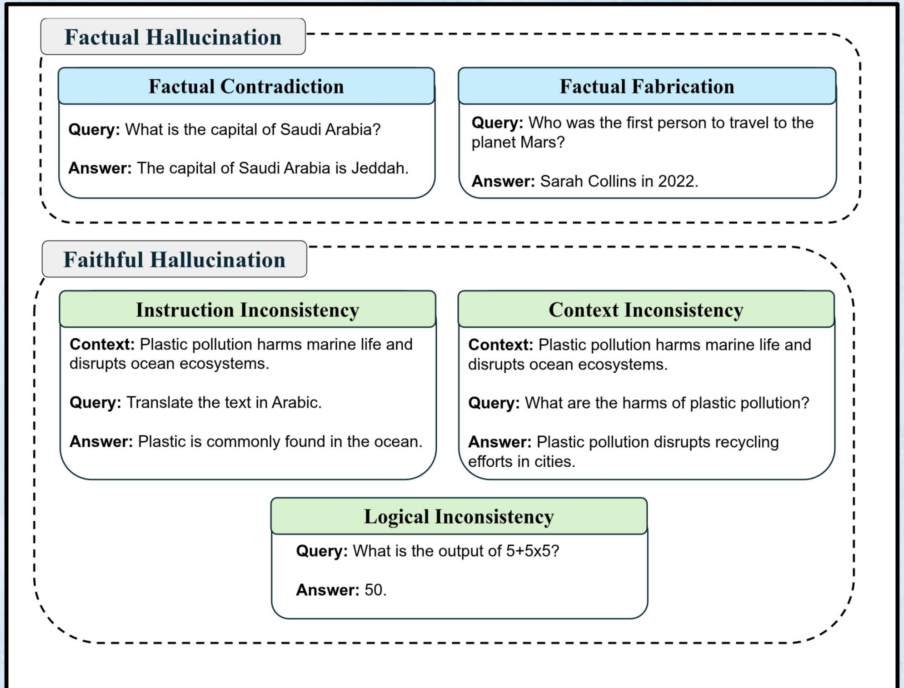
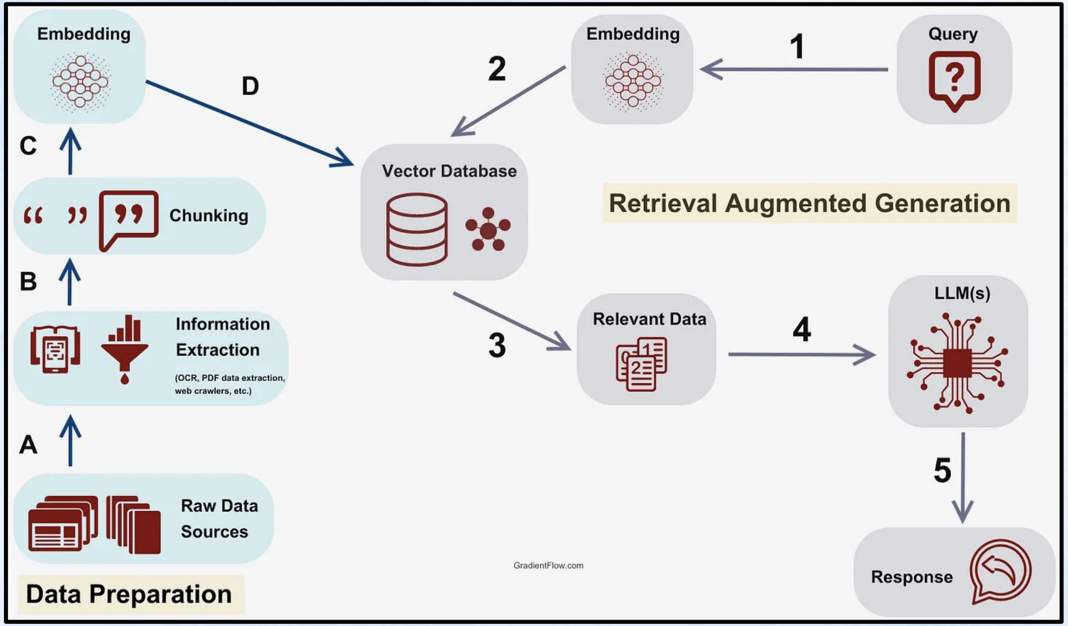


Fig. 3: Examples of various hallucination types in LLM responses.

Alansari, Aisha, and Hamzah Luqman. "Large language models hallucination: A comprehensive survey." Computer Science Review 61 (2026): 100970.

We must ensure *Response Accuracy* of Education Systems

https://gradientflow.substack.com/p/best-practices-in-retrieval-augmented



RAG to the Rescue!

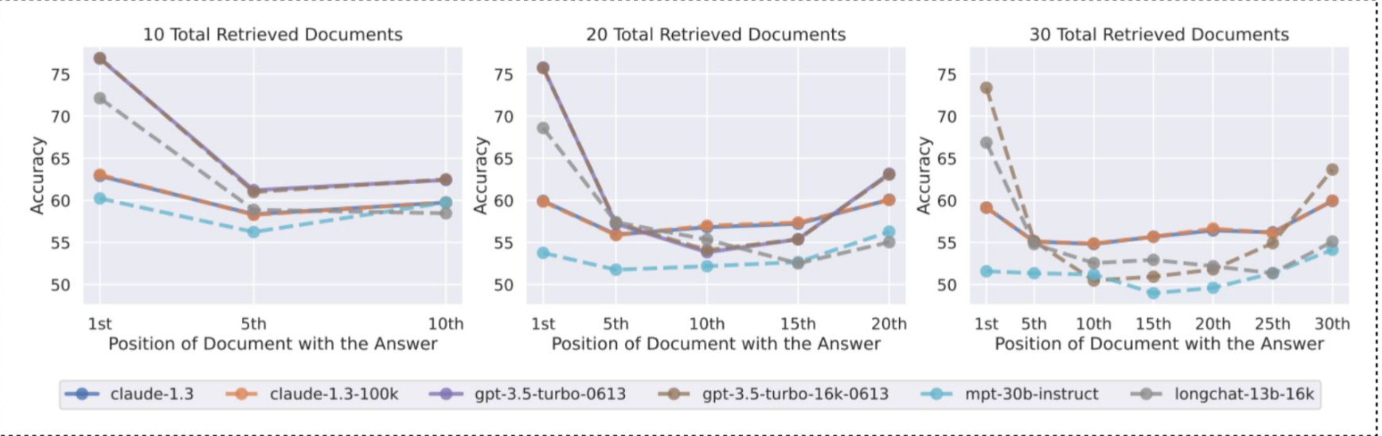
By providing the LLM with true information (retrieved passages), we can **reduce hallucinations and improve accuracy**!

Lewis, Patrick, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." Advances in Neural Information Processing Systems (NeurIPS 2020) 33: 9459–9474.

Why can’t we just feed LLMs everything instead of retrieving?

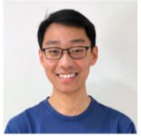
LLM's can't pay attention to the entire context

- The long-context problem bites you – LLMs do not pay attention to its context well!
- Setup: 1 relevant document, all others irrelevant



Best Closed-Book performance: GPT-3.5-Turbo, ~56%
Best Oracle (only feed in relevant doc) performance: GPT-3.5-Turbo, ~88.5%

[Liu+ 2023]



Liu, Nelson F., et al. "Lost in the middle: How language models use long contexts." Transactions of the association for computational linguistics 12 (2024): 157-173.

The retriever is the key component of any RAG system; how to pick your retriever?

Many dimensions

1. Accuracy-style metrics: These will be our focus.
2. Latency: Time to execute a single query.
3. Throughput: Total queries served in a fixed time, perhaps via batch processing.
4. FLOPs: Hardware agnostic measure of compute resources.
5. Disk usage: For the model, index, etc.
6. Memory usage: For the model, index, etc.
7. Cost: Total cost of deployment for a system.

Success and Reciprocal Rank

Rank

For a ranking $\mathbf{D} = [\mathrm{doc}_1, \ldots, \mathrm{doc}_N]$, let

$$\mathbf{Rank}(q, \mathbf{D}) \in \{1, 2, 3, \ldots\}$$

be the position of the **first** relevant document for q in $\mathbf{D}$.

Success

$$\mathrm{Success@K}(q, \mathbf{D}) = \begin{cases} 1 & \text{if } \mathbf{Rank}(q, \mathbf{D}) \leqslant K \\ 0 & \text{otherwise} \end{cases}$$

Reciprocal Rank

$$\mathrm{RR@K}(q, \mathbf{D}) = \begin{cases} \frac{1}{\mathbf{Rank}(q, \mathbf{D})} & \text{if } \mathbf{Rank}(q, \mathbf{D}) \leqslant K \\ 0 & \text{otherwise} \end{cases}$$

MRR@K is the average of this over multiple queries.

Dense Passage Retrieval

- **DENSE representation** retrieval!!
- Uses a dual-encoder dense retriever:
  - one encoder for the question
  - one encoder for the passage
- Trained with positive passages + in-batch / hard negatives to maximize likelihood of predicting positive passage
- Important for RAG because it made neural retrieval **strong enough** to support downstream generation
- Inspired more retrievers, e.g. ColBERT

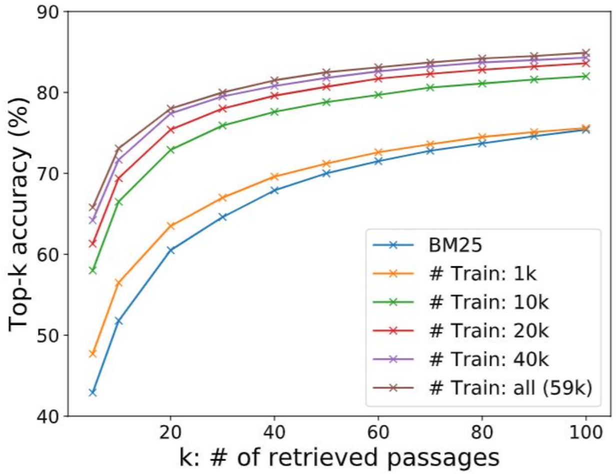
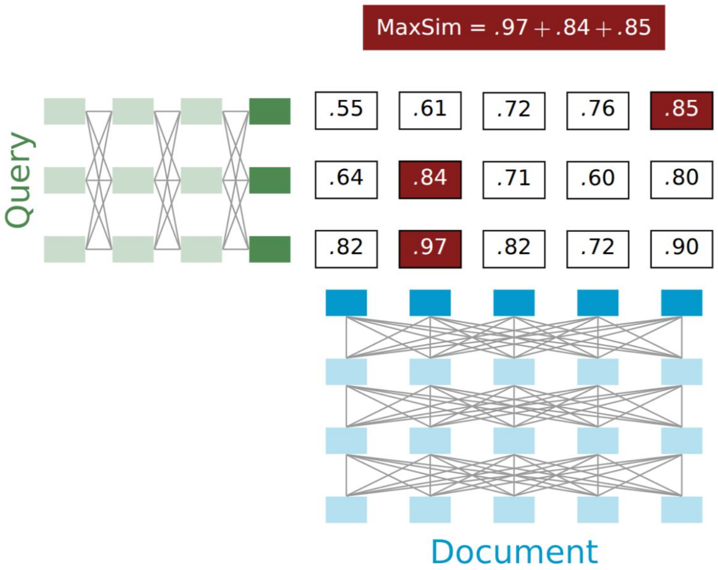


Figure 1: Retriever top-k accuracy with different numbers of training examples used in our dense passage retriever vs BM25. The results are measured on the development set of Natural Questions. Our DPR trained using 1,000 examples already outperforms BM25.

Karpukhin, Vladimir, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. "Dense Passage Retrieval for Open-Domain Question Answering." Proceedings of EMNLP 2020, 6769–6781.

ColBERT



1. Examples: $\langle q_i, \text{doc}_i^+, \{\text{doc}_{i,k}^-\}\rangle$
2. Loss: negative log-likelihood of the positive passage, with **MaxSim** as the basis.

Highly scalable with late, contextual interactions!

For a BERT-style encoder with N layers:

$$\textbf{MaxSim}(q, \text{doc}) = \sum_{i}^{L} \max_{j}^{M} \textbf{Enc}(q)_{N,i}{}^{\top}\textbf{Enc}(\text{doc})_{N,j}$$

with L is the length of q, M the length of doc.

Khattab and Zaharia 2020

- Achieving richness and scalability through **late interactions**!
- Token-wise representations for both queries and passages.

Khattab, Omar, and Matei Zaharia. "Colbert: Efficient and effective passage search via contextualized late interaction over bert." Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval. 2020.

Other Options

- OpenAI's text embeddings
- Sentence Transformers
  - BERT-style models trained to encode semantically similar passages / sentences close together
  - Have models specialized for retrieval
- There are many other neural "information retrieval (IR)" models!

Karpukhin, Vladimir, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. "Dense Passage Retrieval for Open-Domain Question Answering." Proceedings of EMNLP 2020, 6769–6781.

The Canonical RAG Paper *(2020)*

- **Parametric memory** (a pretrained seq2seq model) + **Non-parametric memory** (a retrieved document index)
- Architecture:
  - Retriever gets relevant passages
  - Generator conditions on retrieved passages to produce an answer
- Uses dense retrieval (DPR) over *Wikipedia*
  - <u>*RAG-Sequence:*</u> same retrieved passages condition the whole output
  - <u>*RAG-Token:*</u> retrieved support can vary across generated tokens
- Core motivation: reduce hallucinations and improve knowledge-intensive generation
- Important shift: the model no longer has to store all knowledge in parameters alone

Lewis, Patrick, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." Advances in Neural Information Processing Systems (NeurIPS 2020) 33: 9459–9474.

RAG is a Staple in Interactive Educational Systems

A Large-Scale Real-World Evaluation of an LLM-based VTA

- **Course-specific RAG** over *lecture PDFs, notebooks, and transcribed recordings*
- Materials embedded with text-embedding-3-large into a Faiss DB
- Uses gpt-4o-mini to rewrite dialogue history + latest question into a **context-aware** search query
- Retrieves top-5 chunks, then uses gpt-4o-mini to generate the grounded response
- Evaluation design:
  - 3 rounds of surveys at ***different stages of the course*** to track how student perceptions changed over time
  - Analysis of 3,869 student–VTA interaction pairs
  - Comparison of student–VTA interactions with student–human instructor interactions.

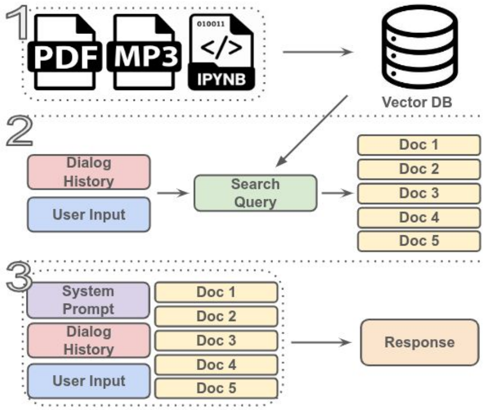


Figure 1: Overview of the VTA architecture. (1) The system processes educational materials into a vector database, (2) retrieves relevant documents based on students' queries, and (3) generates responses.

Kweon, Sunjun, et al. "A large-scale real-world evaluation of an llm-based virtual teaching assistant." Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track). 2025.

JeepyTA

Virtual Teaching Assistants

- **Objective:** Offer round-the-clock assistance to students in an Educational Data Mining course
- **Methods:**
  - Use text embeddings to retrieve **relevant, vetted information** to generate responses
    - **Curated question-answer pairs** from prior iterations of the course
    - Rephrase relevant answer into discussion forum format
    - If top-scoring answer has very low similarity / does not sufficiently address student question, instruct ChatGPT to generate new response
  - Instructor defines scope for moderation, student opt-out feature
  - Notification for instructors when JeepyTA drafts a response, can approve / modify before posting
- **Evaluation & Results:**
  - 15 students filled out survey comparing JeepyTA performance against human TA
    - Generally viewed as on-par as human TAs, but not helpful in areas e.g. useful idea provision
  - Most students still prefer directing questions to human TAs and instructors
  - Comparing against the prior semester: Significant reduction in response time

R. S. Baker, M. Pankiewicz, and X. Liu, “A Step Towards Adaptive Online Learning: Exploring the Role of GPT as Virtual Teaching Assistants in Online Education,” Jul. 31, 2024, EdArXiv. doi: 10.35542/osf.io/rw45b.

RAG: Main Lesson

- To build high-accuracy interactive systems, we need to ground model responses in **vetted, true passages**!
- RAG is really everywhere these days; any responses synthesized through web search or through querying memory can be viewed as RAG
- Education is an especially critical use case for RAG!
    - There are frameworks for making RAG easy, e.g. LangChain, DSPy

04

(Human-Centered) Evaluation

Yay, you built the thing! Now what?

Now that we have powerful LLMs, can they evaluate everything now?

Answer: Not Always!

Human evaluation is still critical!

Especially for nuanced, complex evaluations!!

How do we know what LLMs can vs. cannot evaluate?

Example: Pedagogical Evaluations

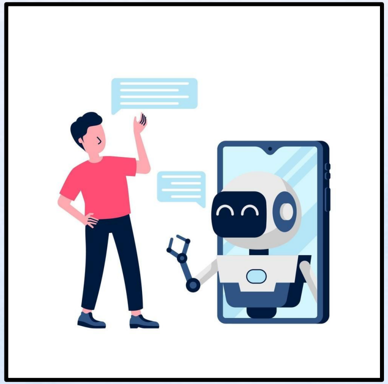
- **Step 1: Build a human gold standard**
  - Define *pedagogy-specific rubric*
  - Collect **expert annotations (at least graduate students in the area)**
  - Verify **inter-rater reliability (e.g. ICC, Cohen's Kappa)**
    - Your results are moot without good reliability!
    - Potentially iterate on the rubric if the reliability is low
- **Step 2: Test the LLM judge against humans**
  - Measure correlation / agreement b/w LLM
  - Check robustness across different conditions (e.g. tasks, response sources, ground truth pedagogical level labels)
- As of recently:
  - Off-the-shelf LLM judges can fail badly on pedagogy-heavy dimensions (Maurya et al.)
  - Expert-labeled data can support more reliable automatic evaluators, but higher-level pedagogy is still hard to judge (Siyan et al.)

[1] Maurya, Kaushal Kumar, et al. "Unifying AI tutor evaluation: An evaluation taxonomy for pedagogical ability assessment of LLM-powered AI tutors." Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers). 2025.
[2] Siyan, Li, et al. "Bringing Pedagogy into Focus: Evaluating Virtual Teaching Assistants' Question-Answering in Asynchronous Learning Environments." arXiv preprint arXiv:2509.17961 (2025).

Types of Human Evaluations

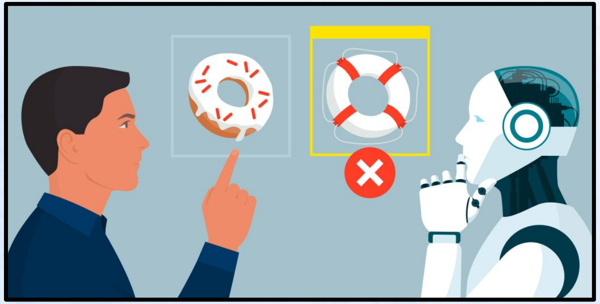
User Study

Within-subject vs. Between-subject designs; one-time vs. longitudinal, etc



Annotations

Rely on expert judgment to guarantee validity



Running User Studies

1. Start with a **clear research question**
2. Choose a *study design*:
   a. Between-subjects: each user sees one system
   b. Within-subjects: each user sees multiple systems
3. Decide what to measure:
   a. Performance / outcomes: learning, task success
   b. Experience: trust, support, workload, satisfaction
   c. Behavior: logs, usage, follow-up actions
4. If you are using self-report measures, try your best to make sure they are validated, standard measures!
   a. If not, you would need to validate them!
5. **Pilot first (small-scale trial run)**, then run the main study

Example: EDEN User Study

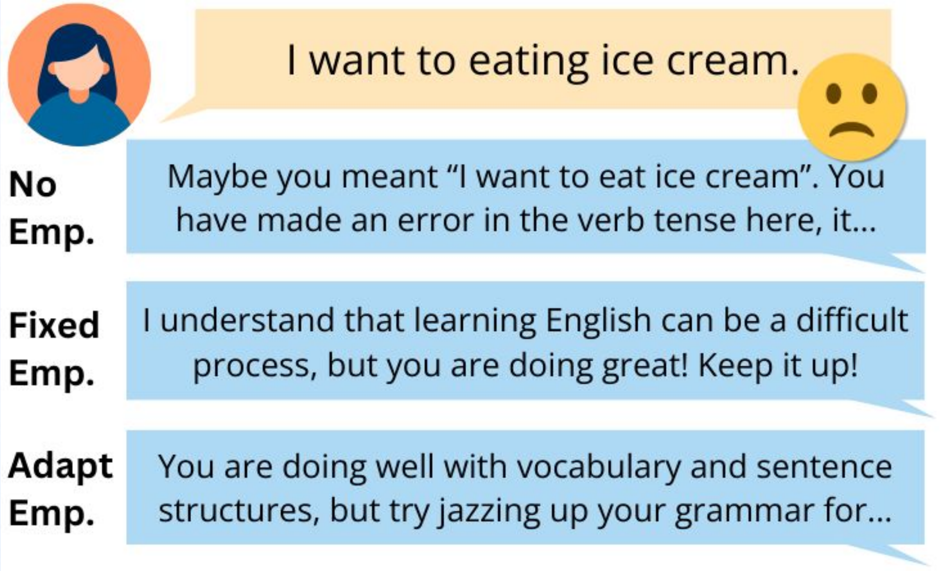
Research Question: Does adaptive, empathetic feedback mechanism improve student L2 Grit by increasing perceived affective support?

Design: Between-Subject Design

Population

- Recruited through social media and personal connections
- **31 Native Mandarin speakers**
- Self-reported English proficiency at least intermediate
- Filled out questionnaire for perceived affective support from chatbot and L2 grit

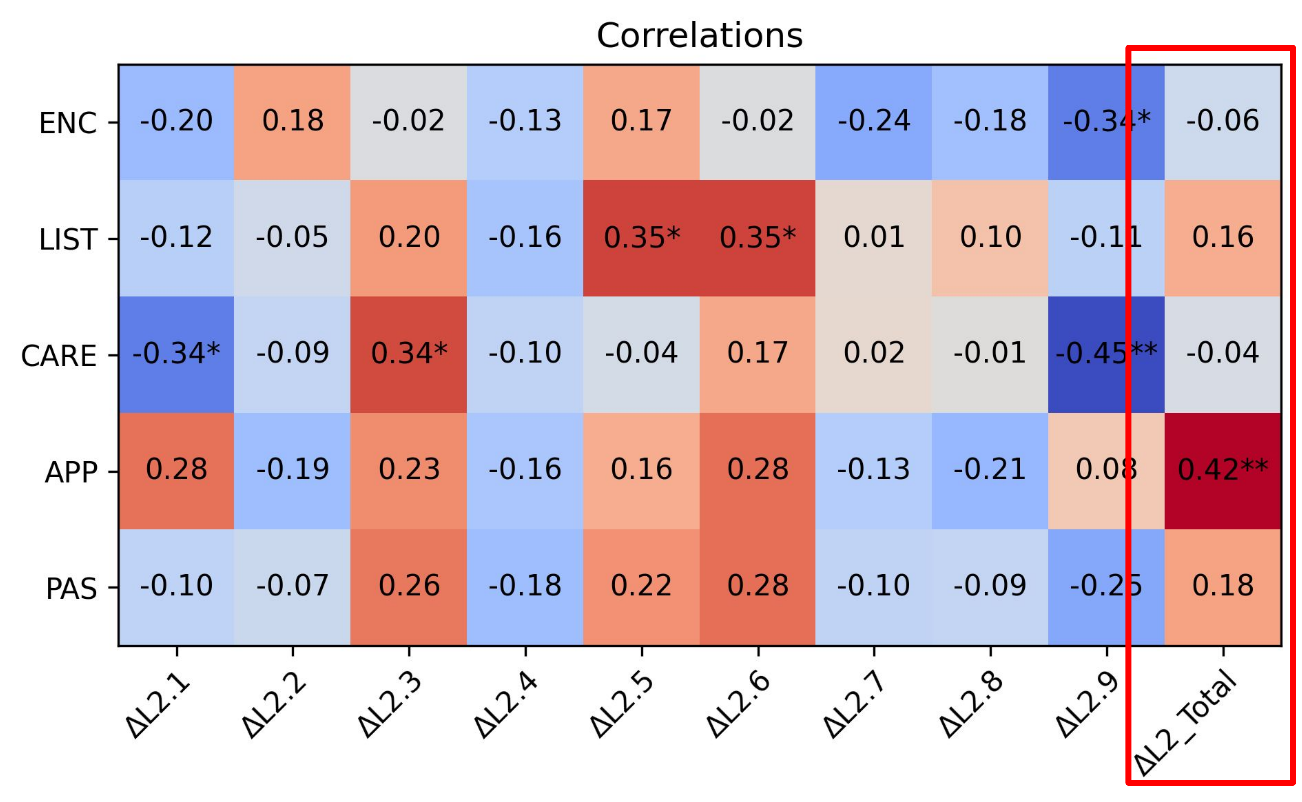
Empathetic Feedback Mechanisms



PAS and Different Empathetic Feedback Mechanisms

	Encourages me when struggling	Listens to me	Cares about my opinion	Appreciates and praises me when doing well	PAS
None	3.53	**4.12**	**4.00**	3.47	3.78
Fixed	3.83	2.83	3.00	3.67	3.33
Adaptive	**4.38**	4.00	3.88	**4.38**	**4.16**

Correlation between PAS and Change in L2 Grit



Human-Centered Evaluation: Main Lesson

- Sometimes, using LLMs for evaluation directly is not sufficient!
  - You can validate LLM judges using annotation-based studies
- Most likely need to run user studies if evaluating interactive systems
- Beware of **confounding factors** in your user study and aim to control confounding factors as much as possible :)
- In education, you would likely need to interact with students and deploy your system in classrooms!

Thank You!

siyan.li@columbia.edu